



UNIVERSIDADE FEDERAL DE OURO PRETO
OFFICE OF THE PROVOST FOR UNDERGRADUATE STUDIES
COURSE SYLLABUS



Course Name in Portuguese: Aprendizado de Máquina Seguro		Code: PCC190	
Course Name in English: Secure Machine Learning			
Department Name and Acronym: Department of Computing - DECOM		Academic Unit: ICEB	
Delivery mode: [x] in-person [] distance			
Semester workload		Weekly workload	
Total 60 hours	Extension 0 hours	Theoretical 4 class hours	Practical 0 class hours
Course description: Fundamentals of machine learning in adversarial environments; taxonomy of attacks and framework for secure learning; causative data poisoning attacks; exploratory evasion and extraction attacks; privacy-preserving mechanisms and differential privacy; attacks and defenses in deep learning; case studies in spam filtering and network security.			
Course content:			
• Introduction and Fundamentals – Machine Learning in Adversarial Contexts – Notation and Review of Statistical Learning Theory – Threat Models and Attack Surfaces • Framework for Secure Learning – Taxonomy of Attacks: Influence, Security Violation, and Specificity – Causative Attacks versus Exploratory Attacks – Secure Learning as a Game between Attacker and Defender – Cost Models and Resilience Evaluation • Causative Attacks (Data Poisoning) – Attacks on Hypersphere-Based Classifiers – Availability Attacks: SpamBayes Case Study – Integrity Attacks: PCA-Based Detector Case Study – Defenses Based on Robust Statistics (ANTIDOTE) • Exploratory Attacks (Evasion and Extraction) – Near-Optimal Evasion of Classifiers – Convexity-Inducing Classifiers and Query Complexity – Model Reverse Engineering and Information Extraction • Privacy in Machine Learning – Differential Privacy: Definitions and Properties – Privacy-Preserving Mechanisms for SVM Learning – Utility and Privacy Bounds • Adversarial Attacks in Deep Learning – Adversarial Examples: FGSM, PGD, and Iterative Methods – Black-Box Attacks and Transferability			



UNIVERSIDADE FEDERAL DE OURO PRETO
OFFICE OF THE PROVOST FOR UNDERGRADUATE STUDIES
COURSE SYLLABUS



- Defenses: Adversarial Training and Defensive Distillation
- Certification and Verifiable Robustness
- Security Case Studies
 - Spam Filtering and Evasion of Email Classifiers
 - Anomaly Detection in Network Traffic
 - Malware Detection and Classifier Evasion
- Future Directions and Open Challenges
 - Theoretical Limits of Adversarial Learning
 - Ethical, Legal, and Public Policy Aspects
 - Adversarial Learning in Large Language Models

Objectives

To provide students with the theoretical and practical foundations needed to analyze, design, and evaluate machine learning systems that operate in adversarial environments. Students are expected to understand the taxonomy of causative and exploratory attacks, apply privacy-preserving mechanisms such as differential privacy, and assess the robustness of deep learning models against adversarial examples, so that they can identify security and privacy risks in real-world machine learning applications and propose adequate defenses.

Methodology

- Lectures: Instructor-led lectures covering the course syllabus.
- Examinations: Individual in-class tests administered during scheduled class sessions.
- Recommended Readings: Reading of technical literature to provide students with opportunities to consult specialized references and to develop their analytical, synthesis, and critical evaluation skills.
- Research Project on Secure Machine Learning.

Evaluation

- One exam worth 3 marks.
- Research project worth 7 marks.

Cronogram

- Week 1 (08/24/2026) – Introduction and Motivation
- Week 2 (08/31/2026) – Background and Notation
- Week 3 (09/07/2026) – Holiday: Brazilian Independence Day
- Week 4 (09/14/2026) – A Framework for Secure Learning: Part I
- Week 5 (09/21/2026) – A Framework for Secure Learning: Part II
- Week 6 (09/28/2026) – Exam I
- Week 7 (10/05/2026) – Attacking a Hypersphere Learner
- Week 8 (10/12/2026) – Holiday: Our Lady of Aparecida Day
- Week 9 (10/19/2026) – Availability Attacks
- Week 10 (10/26/2026) – Integrity Attacks
- Week 11 (11/02/2026) – Holiday: All Souls' Day
- Week 12 (11/09/2026) – Privacy-Preserving Mechanisms
- Week 13 (11/16/2026) – Near-Optimal Evasion of Classifiers



UNIVERSIDADE FEDERAL DE OURO PRETO
OFFICE OF THE PROVOST FOR UNDERGRADUATE STUDIES
COURSE SYLLABUS



- Week 14 (11/23/2026) – Adversarial Machine Learning Challenges
- Week 15 (11/30/2026) – Research Project Development
- Week 16 (12/07/2026) – Research Project Development
- Week 17 (12/14/2026) – Research Project Development
- Week 18 (12/21/2026) – Final Thoughts

Required reading:

- Joseph, A. D.; Nelson, B.; Rubinstein, B. I. P.; Tygar, J. D. Adversarial Machine Learning. Cambridge: Cambridge University Press, 2019.
- Vorobeychik, Y.; Kantarcioglu, M. Adversarial Machine Learning. Synthesis Lectures on Artificial Intelligence and Machine Learning. Morgan & Claypool, 2018.
- Goodfellow, I.; Bengio, Y.; Courville, A. Deep Learning. Cambridge, MA: MIT Press, 2016.

Supplementary reading:

- Dwork, C.; Roth, A. The Algorithmic Foundations of Differential Privacy. Foundations and Trends in Theoretical Computer Science, v. 9, n. 3–4, p. 211–407, 2014.
- Biggio, B.; Roli, F. Wild Patterns: Ten Years After the Rise of Adversarial Machine Learning. Pattern Recognition, v. 84, p. 317–331, 2018.
- Madry, A.; Makelov, A.; Schmidt, L.; Tsipras, D.; Vladu, A. Towards Deep Learning Models Resistant to Adversarial Attacks. In: International Conference on Learning Representations (ICLR), 2018.
- Szegedy, C. et al. Intriguing Properties of Neural Networks. In: International Conference on Learning Representations (ICLR), 2014.
- Papernot, N.; McDaniel, P.; Goodfellow, I.; Jha, S.; Celik, Z. B.; Swami, A. Practical Black-Box Attacks against Machine Learning. In: ACM Asia Conference on Computer and Communications Security (ASIACCS), 2017.